

1 Introduction

Given a book in .txt format, the goal of this assignment is to perform next-character prediction based on fixed-size input window. The dataset used is *Alice's Adventures in Wonderland* by Lewis Carroll.

2 Dataset preparation

The raw text requires preprocessing in order to be used to create the dataset. The pipeline consists of three steps: first, the removal of Project Gutenberg introduction; second, the discard of book's table of contents because it contains non-narrative text; and third, the removal of license section to delete further irrelevant noise.

Before the dataset splits is useful to define two variables: $maxlen = 40$, which specifies the maximum length of the sentences used to train the models; $step = 2$, which specifies the number of skipped characters between two different sentences. After this the text is divided in sentences which are used as dataset. The dataset is splitted in three sets: Train, Test and Validation without shuffling the sentences. *Train set* contains the 80% of the sentences. *Test set* contains $\sim 20\%$ of the remaining sentences. During the splitting process the first $maxlen/skip$ sentences are discarded because the model has already seen this examples during the training phase. A possible overlap is avoided with this strategy. The *Validation set* contains $\sim 20\%$ of Train set instances, the same skip strategy is applied to avoid overlap as well. It is important to note that during the validation phase the data are directly passed to the model. This strategy is used because the text is a temporal series and is useful to maintain this recurrency.

3 Baseline model

The architecture of the baseline model is shown in Table 1.

Layer Type	Configuration	Kernel Size
Conv2D	32 filters	3×3
MaxPooling2D	-	2×2
Conv2D	64 filters	3×3
MaxPooling2D	-	2×2
Flatten	-	-
Dropout	rate = 0.5	-
Dense (Output)	47 units (Softmax)	-

Table 1: Baseline model architecture. The number of output units is the number of different characters inside the book.

The baseline model is a convolutional neural network with 259.503 parameters. After the training and

validation processes during the test phase the model obtains an accuracy of 16,76%.

4 Proposed network

The proposed network combines CNN and LSTM layers. Table 2 shows the proposed architecture which has 202.223 parameters.

Layer Type	Configuration	Kernel Size
Conv1D	128 filters	3×3
Dropout	0.3	-
LSTM	128 cells	-
Dropout	0.4	-
LSTM	64 cells	-
Dropout	0.4	-
Dense (Output)	47 units (Softmax)	-

Table 2: Proposed model architecture. The number of output units is the number of different characters inside the book.

The convolutional layer works as features extractor from the input and tries to find frequent relationships between adjacent characters inside the kernel window. The LSTM layers capture recurrent patterns seen during the training phase and help the architecture to generalize better. The Dropouts layers help the network to avoid overfitting.

The network is trained using: $batch\ size = 2048$, *Adam* as optimizer to avoid an unstable behaviour, $learning\ rate = 0.01$ and $Epochs = 100$ with an early stopping callback with $patience = 3$ to prevent overfitting. This architecture obtains an accuracy of 51,76%.

5 Models evaluation

The two models are compared using Macro-Average ROC curve to show in a more accurate way the quality of the predictions, even for the minority classes.

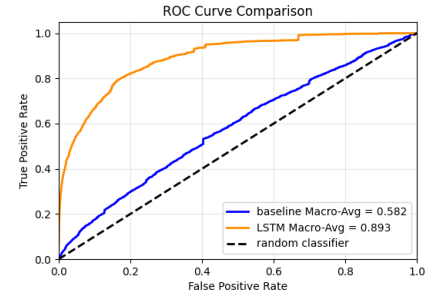


Figure 1: Macro-Average ROC of the two models. LSTM model outperforms the baseline model.

Figure 1 shows that the proposed model outperforms the baseline, demonstrating its effectiveness even on the most infrequent characters.